1.52 Probability: Introduction

Worked Solutions

1 a)

		2nd coin	
		H	T
1st coin	H	HH	HT
	T	TH	TT

1 b)

P(at least one tails) = $\frac{3}{4}$

2 a)

		2nd die			
		1	2	3	4
1st die	1	1	2	3	4
	2	2	4	6	8
	3	3	6	9	12
	4	4	8	12	16

2 b)

P(product > 10) = $\frac{3}{16}$

3 a)

		2nd spinner		
		1	4	9
1st spinner	2	3	6	11
	3	4	7	12
	5	6	9	14
	7	8	11	16

3 b)

P(even) = $\frac{7}{12}$

4 a)

		2nd pack		
		A	B	C
1st pack	A	AA	AB	AC
	B	BA	BB	BC
	C	CA	CB	CC

4 b)

B and C are consonants (while A is a vowel)

P(both consonants) $= \frac{4}{9}$

5 a)

RR, RG, RB, GR, GG, GB, BR, BG

5 b)

P(RR) $= \frac{3}{6} \times \frac{2}{5} = \frac{1}{5}$

P(RG) $= \frac{3}{6} \times \frac{2}{5} = \frac{1}{5}$

P(GR) $= \frac{2}{6} \times \frac{3}{5} = \frac{1}{5}$

P(GG) $= \frac{2}{6} \times \frac{1}{5} = \frac{1}{15}$

P(no blue) $= \frac{1}{5} + \frac{1}{5} + \frac{1}{5} + \frac{1}{15} = \frac{2}{3}$

6 a)

RR, RB, BR, BB

6 b)

P(2 red) $= \frac{3}{6} \times \frac{4}{8} = \frac{1}{4}$

6 c)

P(RB) $= \frac{3}{6} \times \frac{4}{8} = \frac{1}{4}$

P(BR) $= \frac{3}{6} \times \frac{4}{8} = \frac{1}{4}$

P(different colours) $= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

6 d)

P(BR) $= \frac{3}{6} \times \frac{4}{8} = \frac{1}{4}$

P(RB) $= \frac{3}{6} \times \frac{4}{8} = \frac{1}{4}$

Probability $= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

7 a)

		2nd bucket					
		0	1	2	3	4	5
1st bucket	0	00	01	02	03	04	05
	1	10	11	12	13	14	15
	2	20	21	22	23	24	25
	3	30	31	32	33	34	35
	4	40	41	42	43	44	45
	5	50	51	52	53	54	55

7 b)

P(nonzero multiple of 11) $= \frac{5}{36}$

7 c)

P(nonzero multiple of 4) $= \frac{8}{36} = \frac{2}{9}$

8 a)

RRB, RRG, RBR, RBB, RBG, RGR, RGB, BRR, BRB, BRG, BBR, BBG, BGR, BGB, GRR, GRB, GBR, GBB

8 b)

P$(RRB) = \frac{2}{5} \times \frac{1}{4} \times \frac{2}{3} = \frac{4}{60}$
P$(RRG) = \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(RBR) = \frac{2}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{4}{60}$
P$(RGR) = \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(BRR) = \frac{2}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{4}{60}$
P$(GRR) = \frac{1}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{2}{60}$
P(two red) $= \frac{4}{60} + \frac{2}{60} + \frac{4}{60} + \frac{2}{60} + \frac{4}{60} + \frac{2}{60} = \frac{3}{10}$

8 c)

P$(RBG) = \frac{2}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{4}{60}$
P$(RGB) = \frac{2}{5} \times \frac{1}{4} \times \frac{2}{3} = \frac{4}{60}$
P$(BGR) = \frac{2}{5} \times \frac{1}{4} \times \frac{2}{3} = \frac{4}{60}$
P$(BRG) = \frac{2}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{4}{60}$
P$(GRB) = \frac{1}{5} \times \frac{2}{4} \times \frac{2}{3} = \frac{4}{60}$
P$(GBR) = \frac{1}{5} \times \frac{2}{4} \times \frac{2}{3} = \frac{4}{60}$
P(all three colours) $= \frac{4}{60} + \frac{4}{60} + \frac{4}{60} + \frac{4}{60} + \frac{4}{60} + \frac{4}{60} = \frac{2}{5}$

8 d)

P$(RRG) = \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(RGR) = \frac{2}{5} \times \frac{1'}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(GRR) = \frac{1}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(BBG) = \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(BGB) = \frac{2}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{2}{60}$
P$(GBB) = \frac{1}{5} \times \frac{2}{4} \times \frac{1}{3} = \frac{2}{60}$
P(one colour left) $= \frac{2}{60} + \frac{2}{60} + \frac{2}{60} + \frac{2}{60} + \frac{2}{60} + \frac{2}{60} = \frac{1}{5}$

9 a)

P$(A) = 0.25 + 0.3 = 0.55$

9 b)

P$(A \cap B) = 0.3$

9 c)

$\mathrm{P}(B') = 1 - (0.3 + 0.35) = 0.35$

9 d)

$\mathrm{P}(A \cup B) = 0.25 + 0.3 + 0.35 = 0.9$

9 e)

$\mathrm{P}(A \cap B') = 0.25$

9 f)

$\mathrm{P}(A \cup B') = \mathrm{P}(A) + \mathrm{P}(B') - \mathrm{P}(A \cap B')$

$= 0.55 + 0.35 - 0.25 = 0.65$

10 a)

$\mathrm{P}(B) = 0.6 + 0.23 = 0.83$

10 b)

$\mathrm{P}(A' \cap B) = 0.23$

10 c)

$\mathrm{P}(A' \cup B) = \mathrm{P}(A') + \mathrm{P}(B) - \mathrm{P}(A' \cap B)$

$= 0.37 + 0.83 - 0.23 = 0.97$

10 d)

$\mathrm{P}((A \cup B)') = 1 - \mathrm{P}(A \cup B)$

$= 1 - (0.03 + 0.6 + 0.23) = 0.14$

10 e)

$\mathrm{P}((A \cap B)') = 1 - \mathrm{P}(A \cap B) = 1 - 0.6 = 0.4$

10 f)

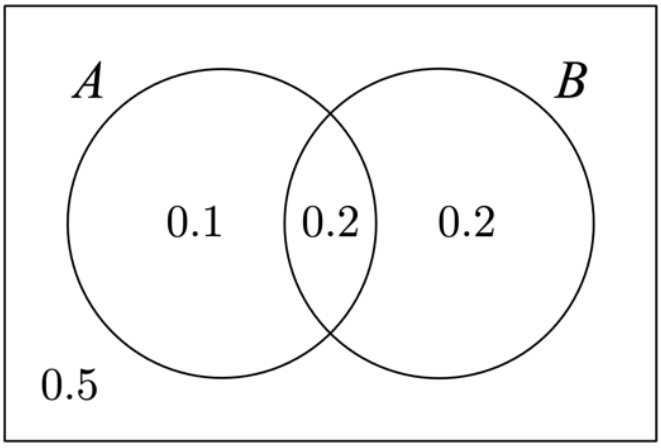
$\mathrm{P}(A' \cup B') = \mathrm{P}(A') + \mathrm{P}(B') - \mathrm{P}(A' \cap B')$

$= 0.37 + 0.17 - 0.14 = 0.4$

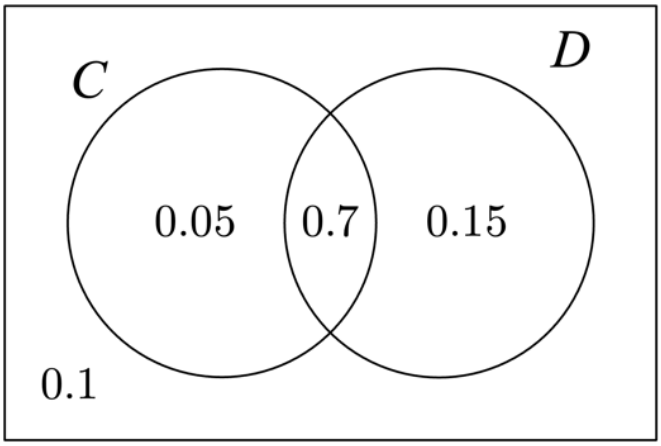
Alternatively:

$\mathrm{P}(A' \cup B') = 1 - \mathrm{P}(A \cap B) = 1 - 0.6 = 0.4$

11)



12)



13 a)

$\frac{1}{5} + k + \frac{1}{3} + k = 1 \Rightarrow 2k = \frac{7}{15} \Rightarrow k = \frac{7}{30}$

13 b)

$\mathrm{P}(A) = \frac{7}{30} + \frac{1}{3} = \frac{17}{30}$

14)

$\frac{1}{6} + p + \frac{2}{5} + q = 1 \Rightarrow p + q = \frac{13}{30} \quad (1)$

$\mathrm{P}(A) = 2\mathrm{P}(B) \Rightarrow p + \frac{2}{5} = 2\left(\frac{2}{5} + q\right)$

$\Rightarrow p + \frac{2}{5} = \frac{4}{5} + 2q \Rightarrow p - 2q = \frac{2}{5} \quad (2)$

$\therefore p = \frac{19}{45}$ and $q = \frac{1}{90}$

15)

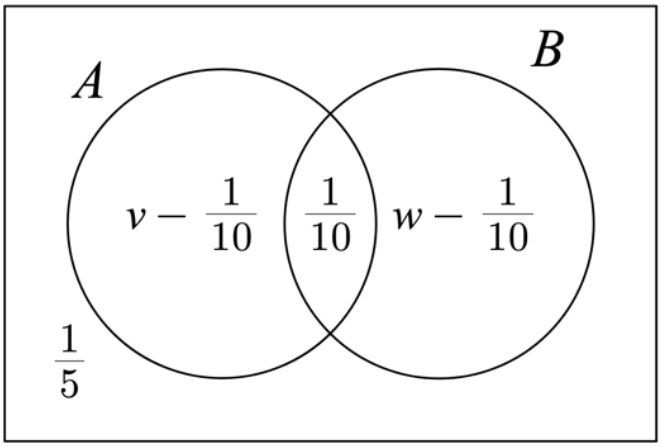
$\frac{2}{9} + r + s + \frac{1}{2} = 1 \Rightarrow r + s = \frac{5}{18} \quad (1)$

$2\mathrm{P}(A) = \mathrm{P}(B) \Rightarrow 2(r + s) = s + \frac{1}{2}$

$\Rightarrow 2r + 2s = s + \frac{1}{2} \Rightarrow 2r + s = \frac{1}{2} \quad (2)$

$\therefore r = \frac{2}{9}$ and $s = \frac{1}{18}$

16 a)



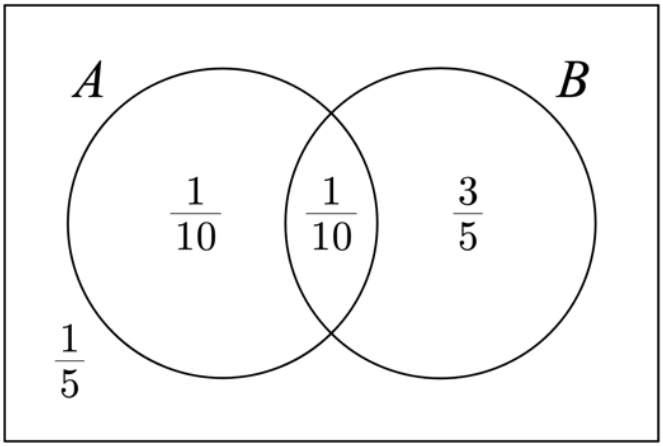
$\frac{1}{5} + v - \frac{1}{10} + \frac{1}{10} + w - \frac{1}{10} = 1 \Rightarrow v + w = \frac{9}{10}$ (1)

$\mathrm{P}(A \cup B) = 4\mathrm{P}(A) \Rightarrow v - \frac{1}{10} + \frac{1}{10} + w - \frac{1}{10} = 4v$

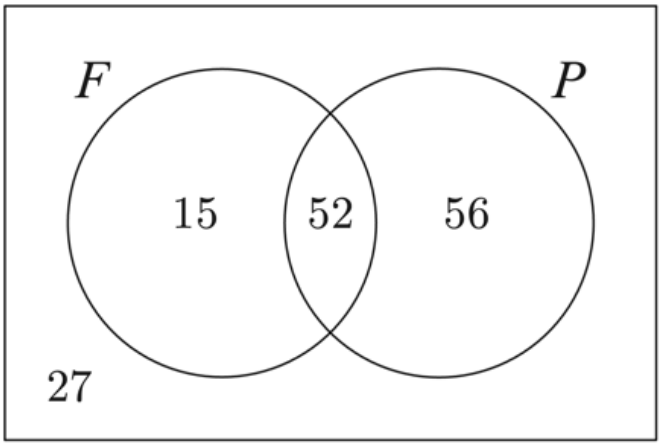
$\Rightarrow 3v - w = -\frac{1}{10}$ (2)

$\therefore v = \frac{1}{5}$ and $w = \frac{7}{10}$

16 b)



17)



18)

$3k + k + 0.2 + k + 0.1 + 0.05 = 1 \Rightarrow 5k = 0.65$

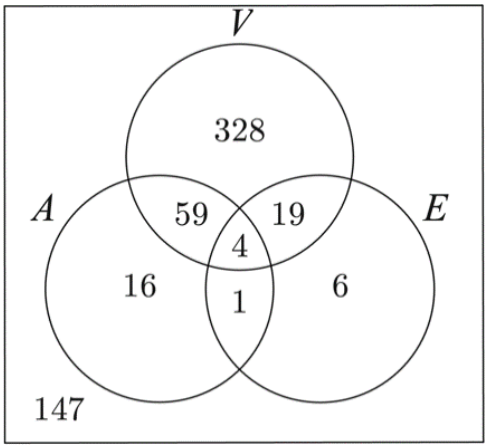
$\Rightarrow k = 0.13$

19)

$0.05 + 0.05 + 0.1 + 0.15 + q + q + q + 0.05 = 1$

$\Rightarrow 3q = 0.6 \Rightarrow q = 0.2$

20)



21 a)

$\mathrm{P}(B) = \frac{1}{6} + \frac{1}{3} + \frac{1}{12} = \frac{7}{12}$

21 b)

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

$= \frac{5}{12} + \frac{7}{12} - \frac{1}{6} = \frac{5}{6}$

21 c)

$\mathrm{P}(B \cup C) = \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(B \cap C)$

$= \frac{7}{12} + \frac{5}{36} - \frac{1}{12} = \frac{23}{36}$

21 d)

$\mathrm{P}(A \cap B) = \frac{1}{6}$

21 e)

$\mathrm{P}(A \cap C) = 0$

21 f)

$\mathrm{P}(A \cup B \cup C) = \frac{1}{4} + \frac{1}{6} + \frac{1}{3} + \frac{1}{12} + \frac{1}{18} = 1 - \frac{1}{9} = \frac{8}{9}$

21 g)

$\mathrm{P}(C') = 1 - \mathrm{P}(C) = 1 - \frac{5}{36} = \frac{31}{36}$

21 h)

$\mathrm{P}(A \cap C') = \mathrm{P}(A) = \frac{5}{12}$

21 i)

$\mathrm{P}(B \cap C') = \frac{1}{6} + \frac{1}{3} = \frac{1}{2}$

21 j)

$\mathrm{P}((A \cup B) \cap C) = \frac{1}{12}$

22 a)

$\mathrm{P}(A) = \frac{3}{50} + \frac{1}{10} + \frac{1}{5} + \frac{1}{50} = \frac{19}{50}$

22 b)

$\mathrm{P}(A \cap B) = \frac{1}{10} + \frac{1}{5} = \frac{3}{10}$

22 c)

$\mathrm{P}(B \cup C) = \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(B \cap C)$

$= \frac{37}{100} + \frac{39}{100} - \frac{11}{50} = \frac{27}{50}$

22 d)

$\mathrm{P}(A' \cap C) = \frac{1}{50} + \frac{3}{20} = \frac{17}{100}$

22 e)

$\mathrm{P}(A \cap B \cap C) = \frac{1}{5}$

22 f)

$\mathrm{P}(A \cup B \cup C) = \frac{1}{20} + \frac{1}{10} + \frac{1}{5} + \frac{1}{50} + \frac{3}{50} + \frac{1}{50} + \frac{3}{20}$

$= 1 - \frac{2}{5} = \frac{3}{5}$

22 g)

$\mathrm{P}(A' \cup B') = \mathrm{P}(A') + \mathrm{P}(B') - \mathrm{P}(A' \cap B')$

$= \frac{31}{50} + \frac{63}{100} - \left(\frac{3}{20} + \frac{2}{5}\right) = \frac{7}{10}$

22 h)

$\mathrm{P}((A \cup B) \cap C) = \frac{1}{50} + \frac{1}{5} + \frac{1}{50} = \frac{6}{25}$

22 i)

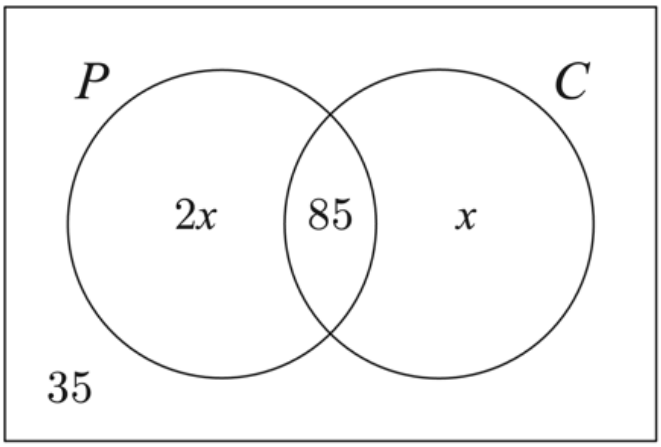
$\mathrm{P}((A \cap B) \cup C) = \mathrm{P}(A \cap B) + \mathrm{P}(C)$

$-\mathrm{P}(A \cap B \cap C) = \frac{3}{10} + \frac{39}{100} - \frac{1}{5} = \frac{49}{100}$

22 j)

$\mathrm{P}(A' \cup B' \cup C') = 1 - \mathrm{P}(A \cap B \cap C) = 1 - \frac{1}{5} = \frac{4}{5}$

23)

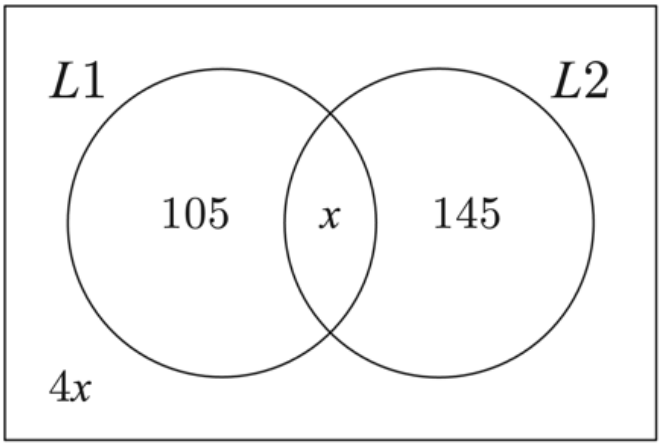


$252 = 35 + 2x + 85 + x \Rightarrow 3x + 120 = 252$

$\Rightarrow x = 44$

$\mathrm{P}(\text{Physics}) = \frac{2\times44+85}{252} = \frac{173}{252} = 0.687$ (3 SF)

24)



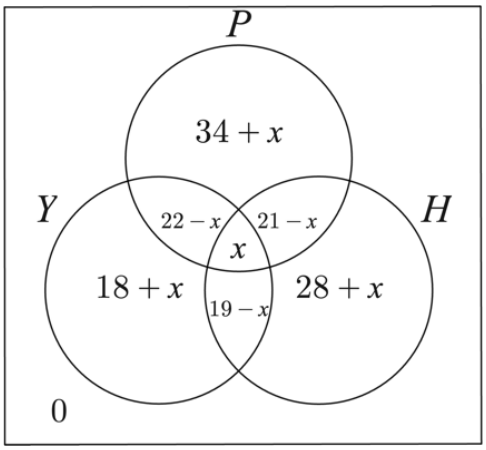
$360 = 4x + 105 + x + 145 \Rightarrow 5x + 250 = 360$

$\Rightarrow x = 22$

P(L2 passkey) $= \frac{22+145}{360} = \frac{167}{360}$

P(Two with a L2 passkey) $= \frac{167}{360} \times \frac{166}{359} = 0.215$ (3 SF)

25)



$$150 = (34 + x) + (22 - x) + x + (21 - x) + (18 + x) + (19 - x) + (28 + x)$$
$$\Rightarrow 150 = 142 + x$$

$\Rightarrow x = 8$

P(all three activities) $= \frac{8}{150} = \frac{4}{75} = 0.0533$ (3 SF)

26 a)

P(both do Art and Book but not Creative Writing)

$= \frac{12}{120} \times \frac{11}{119} = \frac{11}{1190} = 0.00924$ (3 SF)

26 b)

P(1st Creative, 2nd All Three)

$= \frac{38}{120} \times \frac{4}{119} + \frac{4}{120} \times \frac{3}{119} = 0.0115$ (3 SF)

26 c)

P(both do same two clubs)

$= \frac{16}{120} \times \frac{15}{119} + \frac{13}{120} \times \frac{12}{119} + \frac{17}{120} \times \frac{16}{119}$

$= \frac{167}{3570} = 0.0468$ (3 SF)

27 a)

$\mathrm{P}(A) = \frac{60}{150} = \frac{2}{5}$

27 b)

$\mathrm{P}(A \cap B) = \frac{20}{150} = \frac{2}{15}$

27 c)

$\mathrm{P}(B') = \frac{95}{150} = \frac{19}{30}$

27 d)

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

$= \frac{60}{150} + \frac{55}{150} - \frac{20}{150} = \frac{19}{30}$

27 e)

$\mathrm{P}(A \cap B') = \frac{40}{150} = \frac{4}{15}$

27 f)

$\mathrm{P}(A \cup B') = \mathrm{P}(A) + \mathrm{P}(B') - \mathrm{P}(A \cap B')$

$= \frac{60}{150} + \frac{95}{150} - \frac{40}{150} = \frac{23}{30}$

28 a)

$\mathrm{P}(B) = \frac{175}{300} = \frac{7}{12}$

28 b)

$\mathrm{P}(A' \cap B) = \frac{155}{300} = \frac{31}{60}$

28 c)

$\mathrm{P}(A' \cup B) = \mathrm{P}(A') + \mathrm{P}(B) - \mathrm{P}(A' \cap B)$

$= \frac{265}{300} + \frac{175}{300} - \frac{155}{300} = \frac{19}{20}$

28 d)

$\mathrm{P}((A \cup B)') = 1 - \mathrm{P}(A \cup B)$

$= 1 - \left(\frac{35}{300} + \frac{175}{300} - \frac{20}{300}\right) = \frac{11}{30}$

28 e)

$\mathrm{P}((A \cap B)') = 1 - \mathrm{P}(A \cap B) = 1 - \frac{20}{300} = \frac{14}{15}$

28 f)

$\mathrm{P}(A' \cup B') = \mathrm{P}(A') + \mathrm{P}(B') - \mathrm{P}(A' \cap B')$

$= \frac{265}{300} + \frac{125}{300} - \frac{110}{300} = \frac{14}{15}$

29 a i)

$A \cap B'$

The employee travels to and from work by bicycle and has worked at the company for under 5 years.

29 a ii)

$A' \cap B$

The employee does not travel to and from work by bicycle and has worked at the company for over 5 years.

29 a iii)

$A' \cup B'$

The employee does not travel to and from work by bicycle or has worked at the company for under 5 years, or both.

29 b i)

$\mathrm{P}(A) = \frac{2}{19} + \frac{1}{38} = \frac{5}{38}$

29 b ii)

$\mathrm{P}(B) = \frac{2}{19} + \frac{7}{19} = \frac{9}{19}$

29 b iii)

$\mathrm{P}(A \cap B) = \frac{2}{19}$

29 b iv)

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$

$= \frac{5}{38} + \frac{9}{19} - \frac{2}{19} = \frac{1}{2}$

29 b v)

$\mathrm{P}(A' \cap B') = \frac{1}{2}$

29 b vi)

$\mathrm{P}(A' \cup B') = \mathrm{P}(A') + \mathrm{P}(B') - \mathrm{P}(A' \cap B')$

$= \frac{33}{38} + \frac{10}{19} - \frac{1}{2} = \frac{17}{19}$

30 a)

$\mathrm{P}((A' \cup B)') = 1 - \left(\frac{63}{100} + \frac{59}{100} - \frac{36}{100}\right) = \frac{7}{50} = 14\%$

30 b)

$\mathrm{P}((A \cap B')') = 1 - \frac{14}{100} = \frac{43}{50} = 86\%$

31)

$\mathrm{P}(A) = 3\mathrm{P}(B) \Rightarrow 0.25 + 0.41 = 3\mathrm{P}(B)$

$\Rightarrow \mathrm{P}(B) = 0.22$

$\therefore \mathrm{P}(A' \cap B) = 0.22 - 0.19 = 0.03$

$\therefore \mathrm{P}(A' \cap B') = 1 - (0.19 + 0.47 + 0.03) = 0.31$

	A	A'
B	0.19	0.03
B'	0.47	0.31

32 a)

$\mathrm{P}(C \cap D') = 0.05$

32 b)

$\mathrm{P}(B \cup D) = \mathrm{P}(B) + \mathrm{P}(D) - \mathrm{P}(B \cap D)$

$= 0.35 + 0.65 - 0.25 = 0.75$

32 c)

$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) = 0.3 + 0.35 = 0.65$

32 d)

$\mathrm{P}((A \cup C) \cap D') = 0.2 + 0.05 = 0.25$

32 e)

$\mathrm{P}((A \cap D) \cup B) = \mathrm{P}(A \cap D) + \mathrm{P}(B) - \mathrm{P}(A \cap D \cap B)$

$= 0.1 + 0.35 - 0 = 0.45$

33)

$0.021 = 0.4 \times 0.015 + 0.6 \times x \Rightarrow x = 0.025$

So 2.5% of the onions bought from farm B are spoilt.

34)

$0.8125 = 0.85 \times 0.25 + 0.7 \times 0.45 + 0.3 \times x$

$\Rightarrow 0.3x = 0.285 \Rightarrow x = 0.95$

So 95% of the students from class C are right-handed.

35)

$p + 3p + q + 5q = 1 \Rightarrow 4p + 6q = 1 \quad (1)$

$3(p + q) = 5(p + 3p) \Rightarrow 3p + 3q = 20p$

$\Rightarrow 17p - 3q = 0 \quad (2)$

$\therefore p = \frac{1}{38}$ and $q = \frac{17}{114}$

36)

$2r + t + 3s + 3r + 2t + s = 1$

$\Rightarrow 5r + 4s + 3t = 1 \quad (1)$

$2r + t = 0.26 \quad (2)$

$3r + s + t = 0.44 \quad (3)$

$\therefore r = 0.1,\ s = 0.08,\ t = 0.06$

37 a)

$\mathrm{P}(A) = \frac{1}{3}$

37 b)

$\mathrm{P}(A') = \frac{2}{3}$

37 c)

$\mathrm{P}(B) = \frac{1}{3} \times \frac{1}{4} + \frac{2}{3} \times \frac{3}{5} = \frac{29}{60}$

37 d)

$\mathrm{P}(B') = 1 - \frac{29}{60} = \frac{31}{60}$

38 a)

$\mathrm{P}(A \text{ then } B \text{ then } C) = \frac{1}{2} \times \frac{1}{5} \times \frac{1}{6} = \frac{1}{60}$

38 b)

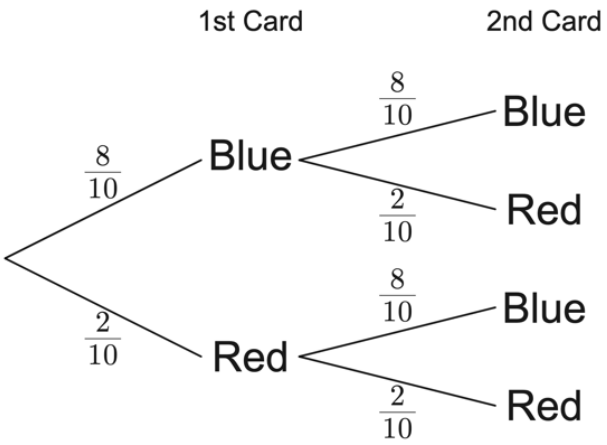
$\mathrm{P}(A' \text{ then } B \text{ then } C') = \frac{1}{2} \times \frac{3}{5} \times \frac{1}{5} = \frac{3}{50}$

38 c)

$\mathrm{P}(C) = \frac{1}{2} \times \frac{1}{5} \times \frac{1}{6} + \frac{1}{2} \times \frac{4}{5} \times \frac{1}{4} + \frac{1}{2} \times \frac{3}{5} \times \frac{4}{5} + \frac{1}{2} \times \frac{2}{5} \times \frac{3}{5}$

$= \frac{1}{60} + \frac{1}{10} + \frac{6}{25} + \frac{3}{25} = \frac{143}{300}$

39 a)



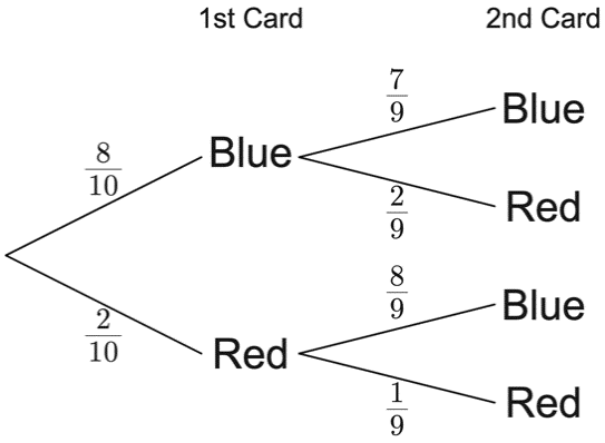
39 b)

P(Same colour)

$= \text{P(Blue then Blue)} + \text{P(Red then Red)}$

$= \left(\frac{8}{10}\right)^2 + \left(\frac{2}{10}\right)^2 = \frac{17}{25}$

40 a)



40 b)

P(Same colour)

$= \text{P(Blue then Blue)} + \text{P(Red then Red)}$

$= \frac{8}{10} \times \frac{7}{9} + \frac{2}{10} \times \frac{1}{9} = \frac{29}{45}$

41 a)

$\text{P}(B \text{ then } D) = \frac{1}{8} \times \frac{1}{3} = \frac{1}{24}$

41 b)

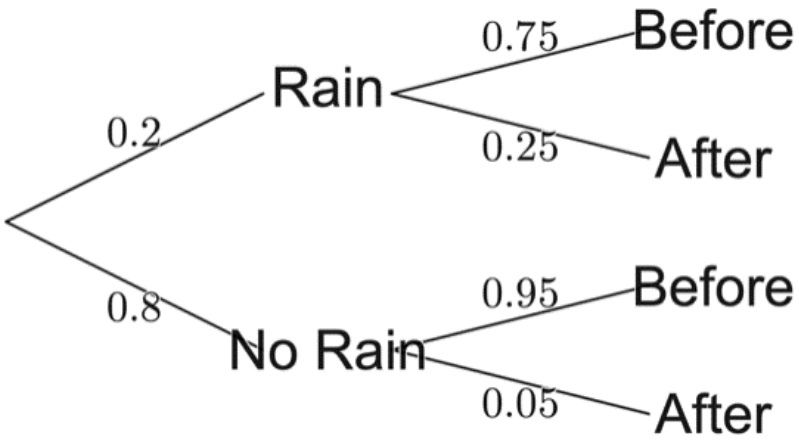
$\text{P}(E) = \frac{5}{8} \times \frac{4}{9} + \frac{1}{8} \times \frac{2}{3} + \frac{1}{4} \times \frac{1}{6} = \frac{5}{18} + \frac{1}{12} + \frac{1}{24} = \frac{29}{72}$

41 c)

$\text{P}(A \text{ or } E \text{ or both}) = \frac{5}{8} + \frac{1}{8} \times \frac{2}{3} + \frac{1}{4} \times \frac{1}{6}$

$= \frac{5}{8} + \frac{1}{12} + \frac{1}{24} = \frac{3}{4}$

42 a)



42 b)

P(Arrives after 8:30am) $= 0.2 \times 0.25 + 0.8 \times 0.05$

$= 0.09$

42 c)

P(Arrives before 8:30am) $= 1 - 0.09 = 0.91$

P(Arrives after 8:30 on all five days) $= 0.91^5$

$= 0.624$ (3 SF)

43 a)

P(both aged 65 or over) $= 0.1917^2 = 0.0367$ (3 SF)

43 b)

P(65 or over and 14 or under)

$= 0.1917 \times 0.1747 + 0.1747 \times 0.1917$

$= 0.0670$ (3 SF)

44 a)

Total $= 46 + 194 + 140 = 380$

P(both aged 65 or over) $= \frac{140}{380} \times \frac{139}{379} = \frac{973}{7201}$

$= 0.135$ (3 SF)

44 b)

P(65 or over and 14 or under)

$= \frac{140}{380} \times \frac{46}{379} + \frac{46}{380} \times \frac{140}{379} = \frac{644}{7201} = 0.089$ (3 SF)

45 a)

$\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4} \times \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{720}$

45 b)

Then the probability of picking any card other than 5 or 6 is $\frac{4}{6}$, then $\frac{3}{5}$, then $\frac{2}{4}$, and then $\frac{1}{3}$. Then there are two cards left and we want 5 and then 6.

$\frac{4}{6} \times \frac{3}{5} \times \frac{2}{4} \times \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{30}$

45 c)

The probability of picking 1 first is $\frac{1}{6}$

Then the probability of picking any card other than 6 is $\frac{4}{5}$, then $\frac{3}{4}$, then $\frac{2}{3}$, and then $\frac{1}{2}$. Then there is only one card left, the 6.

$\frac{1}{6} \times \frac{4}{5} \times \frac{3}{4} \times \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{1}{30}$

45 d)

The first card can be 1, 3 or 5, so the probability of this happening is $\frac{3}{6} = \frac{1}{2}$

Then the probability of the second card being 2 is $\frac{1}{5}$

Then the third card can be one of the two 1, 3 or 5 that is left, so the probability of this happening is $\frac{2}{4} = \frac{1}{2}$

Then the probability of the fourth card being 4 is $\frac{1}{3}$

Then the fifth card must be the one remaining card of 1, 3 or 5, so the probability of this happening is $\frac{1}{2}$

Then there is only one card left, the 6.

$\frac{3}{6} \times \frac{1}{5} \times \frac{2}{4} \times \frac{1}{3} \times \frac{1}{2} \times 1 = \frac{1}{120}$

46 a)

$\frac{3}{8} \times \frac{2}{7} \times \frac{1}{6} = \frac{1}{56}$

46 b)

$\frac{5}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{2}{5} \times \frac{1}{4} = \frac{1}{56}$

46 c)

$\frac{5}{8} \times \frac{3}{7} \times \frac{4}{6} \times \frac{2}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{56}$

47 a)

1	2	3	4	5	6	7	8	9	10
	L	L	W	L		L	W		

There are four prime numbers between 1 and 10: 2, 3, 5 and 7. Any of these being picked will lose the game.

$\frac{4}{10} = \frac{2}{5}$

47 b)

Pete must pick a card that neither wins or loses the game. So he must not pick 2, 3, 4, 5, 7 or 8. Then Harvey must pick 4 or 8.

$\frac{4}{10} \times \frac{2}{9} = \frac{4}{45}$

47 c)

$\frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} = \frac{1}{210}$

47 d)

Either Pete picks a multiple of 4 with his first card, or Pete picks a card with no outcome and Harvey picks a prime.

$\frac{2}{10} + \frac{4}{10} \times \frac{4}{9} = \frac{17}{45}$

47 e)

Either Pete wins with his first card, or Harvey loses with his first card, or Pete wins with his second card, or Harvey loses with his second card, or Pete wins with his third card.

$$\begin{aligned} &\frac{2}{10} + \frac{4}{10} \times \frac{4}{9} + \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} + \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{4}{7} \\ &\quad + \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} \times \frac{2}{6} = \frac{1}{5} + \frac{8}{45} + \frac{1}{30} + \frac{2}{105} + \frac{1}{630} \\ &= \frac{136}{315} = 0.432 \quad (3 \text{ SF}) \end{aligned}$$

47 f)

Either Pete loses with his first card, or Harvey wins with his first card, or Pete loses with his second card, or Harvey wins with his second card, or Pete loses with his third card.

$$\begin{aligned} &\frac{4}{10} + \frac{4}{10} \times \frac{2}{9} + \frac{4}{10} \times \frac{3}{9} \times \frac{4}{8} + \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{2}{7} \\ &\quad + \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} \times \frac{4}{6} = \frac{2}{5} + \frac{4}{45} + \frac{1}{15} + \frac{1}{105} + \frac{1}{315} \\ &= \frac{179}{315} = 0.568 \quad (3 \text{ SF}) \end{aligned}$$

48)

$\mathrm{P}(\text{1st and Last both Red}) = \mathrm{P}(RBRR) + \mathrm{P}(RRBR)$

$= \frac{3}{4} \times \frac{1}{3} \times 1 + \frac{3}{4} \times \frac{2}{3} \times \frac{1}{2} \times 1 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

49)

$\mathrm{P}(RBR) + \mathrm{P}(BRR) = \frac{2}{3} \times \frac{1}{2} \times 1 + \frac{1}{3} \times 1 = \frac{2}{3}$

50 a)

$\mathrm{P}(\text{Same Colour}) = \mathrm{P}(RR) + \mathrm{P}(BB)$

$= \frac{3}{18} \times \frac{4}{9} + \frac{10}{18} \times \frac{5}{9} = \frac{31}{81}$

50 b)

Option 1: Bag A gains a red counter.

Bag A now contains 4 red, 10 blue, 5 yellow

Bag B now contains 3 red, 5 blue

$\mathrm{P}(\text{Same Colour}) = \mathrm{P}(RR) + \mathrm{P}(BB)$

$= \frac{4}{19} \times \frac{3}{8} + \frac{10}{19} \times \frac{5}{8} = \frac{31}{76}$

Option 2: Bag A gains a blue counter.

Bag A now contains 3 red, 11 blue, 5 yellow

Bag B now contains 4 red, 4 blue

$\mathrm{P}(\text{Same Colour}) = \mathrm{P}(RR) + \mathrm{P}(BB)$

$= \frac{3}{19} \times \frac{4}{8} + \frac{11}{19} \times \frac{4}{8} = \frac{7}{19}$

$\therefore \mathrm{P}(\text{Same Colour}) = \frac{31}{76} + \frac{7}{19} = \frac{59}{76}$

51)

$\mathrm{P}(\text{Different Colour}) = \frac{7}{9}$

$\Rightarrow 1 - \mathrm{P}(\text{Same Colour}) = \frac{7}{9}$

$\Rightarrow 1 - (\mathrm{P}(RR) + \mathrm{P}(YY) + \mathrm{P}(GG)) = \frac{7}{9}$

$\Rightarrow 1 - \left(\frac{8}{18} \times \frac{3}{5+k} + \frac{7}{18} \times \frac{2}{5+k} + \frac{3}{18} \times \frac{k}{5+k}\right) = \frac{7}{9}$

$\Rightarrow 1 - \frac{38+3k}{18(5+k)} = \frac{7}{9} \Rightarrow \frac{38+3k}{18(5+k)} = \frac{2}{9}$

$\Rightarrow \frac{38+3k}{2(5+k)} = 2$

$\Rightarrow 38 + 3k = 20 + 4k$

$\Rightarrow k = 18$

52)

Bag A contains 4 red, 5 yellow, 3 green

Bag B contains 3 red, 2 yellow, 1 green

$\mathrm{P}(RR) + \mathrm{P}(YY) + \mathrm{P}(GG) = \frac{3}{6} \times \frac{5}{13} + \frac{2}{6} \times \frac{6}{13} + \frac{1}{6} \times \frac{4}{13}$

$= \frac{31}{78}$

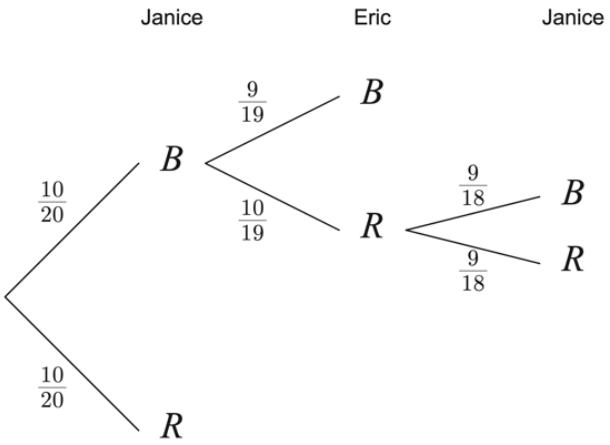
53)

Bag A contains 6 red, 3 yellow, 2 green

Bag B contains 5 red, 4 yellow, 2 green

$\mathrm{P}(RG) + \mathrm{P}(GY) = \frac{5}{11} \times \frac{2}{12} + \frac{2}{11} \times \frac{3}{12} = \frac{4}{33}$

54 a)



54 b)

P(6 marbles) $= \frac{10}{20} \times \frac{10}{19} \times \frac{9}{18} \times \frac{9}{17} \times \frac{8}{16} \times \frac{8}{15} = \frac{6}{323}$

55 a)

P(1 Complex Loci, 1 Matrices, 1 Series)

$= \mathrm{P}(CMS) + \mathrm{P}(CSM) + \mathrm{P}(MCS) + \mathrm{P}(MSC) + \mathrm{P}(CMS) + \mathrm{P}(CSM)$

$= 3! \times \frac{20}{48} \times \frac{18}{47} \times \frac{10}{46} = \frac{225}{1081}$

55 b)

P(3 Complex Loci) $= \frac{20}{48} \times \frac{19}{47} \times \frac{18}{46} = \frac{285}{4324}$

55 c)

P(one student from each class)

$= \mathrm{P}(ABC) + \mathrm{P}(ACB) + \mathrm{P}(BAC) + \mathrm{P}(BCA) + \mathrm{P}(CAB) + \mathrm{P}(CBA)$

$= 6! \times \frac{15}{48} \times \frac{16}{47} \times \frac{17}{46} = \frac{255}{1081}$

55 d)

P(2 Matrices, 1 Series)

$= \mathrm{P}(MMS) + \mathrm{P}(MSM) + \mathrm{P}(SMM)$

$= 3 \times \frac{18}{48} \times \frac{17}{47} \times \frac{10}{46} = \frac{765}{8648}$

56 a)

P(Erin wins) $= 0.75 \times 0.75 + 0.75 \times 0.25 \times 0.75 \times 0.75 + 0.25 \times 0.75 \times 0.75$

$= 0.809$ (3 SF)

56 b)

P(neither wins) $= 0.75 \times 0.25 \times 0.75 \times 0.25 + 0.25 \times 0.75 \times 0.25 \times 0.75$

$= 0.0703$ (3 SF)

57 a)

Let CS mean Cecil Scores and CM mean Cecil Misses

Let HS mean Hilda Scores and HM mean Hilda Misses

Either:

CS, CS, CS

and

HS, HS, HM

or HS, HM, HS

or HS, HM, HM

or HM, HS, HS

or HM, HS, HM

or HM, HM, HS

or HM, HM, HM

$= 0.35^3 \times \left((3 \times 0.55^2 \times 0.45) + (3 \times 0.55 \times 0.45^2) + 0.45^3\right) = 0.03574167188\ldots$

Or:

CS, CS, CM

or CS, CM, CS

or CM, CS, CS

and

HS, HM, HM

or HM, HS, HM

or HM, HM, HS

or HM, HM, HM

$= (3 \times 0.35^2 \times 0.65) \times \left((3 \times 0.55 \times 0.45^2) + 0.45^3\right) = 0.1015815938\ldots$

So P(Cecil Wins) $= 0.0357\ldots + 0.1015\ldots = 0.137$ (3 SF)

57 b)

CS, CS, CS

and

HS, HS, HS

$= 0.35^3 \times 0.55^3 = 0.007133328125\ldots$

Or:

CS, CS, CM

or CS, CM, CS

or CM, CS, CS

and

HS, HS, HM

or HS, HM, HS

or HM, HS, HS

$= (3 \times 0.35^2 \times 0.65) \times (3 \times 0.55^2 \times 0.45) = 0.09755057813\ldots$

Or:

CS, CM, CM

or CM, CS, CM

or CM, CM, CS

or CM, CM, CM

and

HS, HM, HM

or HM, HS, HM

or HM, HM, HS

or HM, HM, HM

$= (3 \times 0.35 \times 0.65^2 + 0.65^3) \times (3 \times 0.55 \times 0.45^2 + 0.45^3) = 0.3054358125\ldots$

P(Both Lose) $= 0.0071\ldots + 0.0975\ldots + 0.3054\ldots = 0.410$ (3 SF)